

Finite-Agent Coordination on Cell-Truth: Calculus for Purpose, Belief Incompatibility, and the Reduction of Meaning to Cell-Membership

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Abstract

This monograph develops a comprehensive mathematical theory of multi-agent coordination grounded in a single primitive: the receiver-relative scalar S measuring residual semantic distance to an action-cell. The principal results are organized into seven theorem clusters. (I) *Cell-Truth foundations*: a self-contained recapitulation of the receiver calculus, the receiver floor $\beta > 0$, the action-cell with positive tolerance, and representational invariance. (II) *Multi-agent algebra*: the formal extension of the calculus to n agents with distinct knowledge frameworks, decoders, motivations, and methodologies; we prove a multi-agent catalytic composition law and a reachability monotonicity theorem. (III) *Common-Cell Convergence*: agents with mutually disjoint representation spaces and incompatible belief systems can nevertheless attain $S \leq \tau(\mathcal{C})$ on the same action-cell; we exhibit explicit constructions and prove a reachability lower bound. (IV) *Purpose as Fixed-Point*: a purpose is formalized as a fixed-point cell of the multi-agent composite floor functional; we prove existence, uniqueness up to reachability equivalence, and a stability theorem under agent perturbation. (V) *Motivation Heterogeneity*: the composite floor depends only on the methodology floors $S_b(\mathfrak{P}_i)$, not on the methodologies' internal goal-content; agents running entirely different objective functions can converge on a common cell. (VI) *Cell-Meaning replaces Point-Meaning*: we reformulate the classical

“impossibility of meaning” problem as a category error (seeking point-meaning where only cell-meaning exists), and show that all eleven classical meaning-prerequisites collapse to the single condition $\beta = 0$, which is forbidden by the floor theorem. (VII) *Knowledge-Understanding gap and bias-as-decoder*: the classical Gödelian residue between accessing a solution and deriving it is exactly the receiver floor β ; the so-called observation-bias equivalence becomes the decoder-projection composition. Forty-five numerical experiments verify all claims to machine precision; results are summarized in Section 18 and visualized in five publication panels. The calculus is independent and self-contained.

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Part I

Foundations

1 Introduction

1.1 The coordination puzzle

Consider three observers at a zoo confronting a confined large cat. Observer A has never seen a feline before; observer B is a trained biologist; observer C is a child accompanied by a parent. Each operates with a different cognitive substrate: different concepts, different representations, different decision-criteria, different motivations. Yet, with high reliability, all three achieve the same outcome: they keep a safe distance.

This empirical regularity is striking. The three observers do not share beliefs (the child does not believe “this is *Panthera leo*”; the naïve observer does not have a concept “cat”; the biologist’s beliefs are saturated with phylogenetic detail). They do not share concepts. They do not even share natural-language vocabulary (substitute three speakers of mutually unintelligible languages and the coordination still holds). What they share is the *action*: each ends up in the distance-keeping region of the outcome space.

Generalize. Multi-agent coordination on common goals appears reliably across:

- physical safety (driving conventions, collision avoidance);
- economic exchange (price discovery, market clearing);

- scientific replication (independent experiments converging on the same finding through different equipment, different protocols, different instruments);
- social institutions (linguistic conventions, traffic signals, queueing behaviour);
- collective construction (multi-trade construction projects, distributed software development).

In each case agents with markedly different internal states converge on markedly similar outcomes. The standard explanation, going back to Lewis [35] and Schelling [42], is that coordination is sustained by *common knowledge* or *focal points*. But common knowledge is an extreme idealization [4, 18]: it requires not just that all agents know X , but that all know that all know X , and so on. For agents with mutually inaccessible knowledge frameworks (different species, different levels of expertise, different languages, different cognitive architectures), common knowledge is unreachable. Yet coordination persists.

This monograph offers a different account. The coordination object is not in the agents’ representation spaces at all. It is in the outcome space. The agents’ representations need not agree — they only need to *decode to overlapping action-cells*.

1.2 Cell-truth as the coordination substrate

Our previous work (Section 2 below recapitulates the calculus self-containedly) established the *Cell-Truth Theorem*: the operational truth-bearer of bounded inquiry is a positive-tolerance region $\mathcal{C} \subseteq \mathcal{X}$ in outcome space, not a point in any receiver’s representation $\mathcal{K}_i$. This region is the *action-equivalence cell*: the set of states that, under the receiver’s action map, produce indistinguishable practical outputs.

The present monograph develops the consequences of cell-truth for multi-agent coordination. The principal claim is:

Coordination among n finite agents on a common goal is mathematically equivalent to the existence of an action-cell $\mathcal{C} \subseteq \mathcal{X}$ such that every

agent's decoder–projection chain admits a candidate at distance $\leq \tau(\mathcal{C}) - \beta_i$ from $\mathcal{C}$, regardless of whether the agents' representation spaces, beliefs, motivations, or methodologies share any common content.

We prove this as the *Common-Cell Convergence Theorem* (Theorem 5.1) and develop its consequences. The architecture so obtained dispenses entirely with shared knowledge, shared belief, shared concepts, or shared motivation as prerequisites for coordination. What replaces them is the *cell itself* as a structure in outcome space, and the agents' multiplicative catalytic composition law as the mechanism by which finer cells become reachable as more agents participate.

1.3 What this manuscript replaces

Several classical and recent literatures have struggled with the question of how meaning, purpose, and coordination can survive the strict computational and information-theoretic constraints on finite observers. Three threads recur:

(1) *The impossibility of meaning.* A standard catalogue of prerequisites for meaningful systems — access to predetermined temporal coordinates, absolute coordinate precision, oscillatory convergence control, indefinite quantum coherence, consciousness substrate independence, collective truth verification, thermodynamic reversibility, problem-solution method determinability, zero temporal delay of understanding, information conservation, temporal dimension fundamentality — contains items that are individually impossible under standard physics. The classical conclusion is that meaning is impossible.

(2) *The Gödelian residue.* For solutions to nontrivial problems, the classical knowledge–understanding gap creates irreducible unknowables: even when a solution is accessible, why *this* solution rather than another, and how it arose, remains undeterminable [21, 15]. The classical conclusion is that finite observers operate with persistent residues of unknowable content.

(3) *The observation–bias equivalence.* Functional observation by finite observers requires

selection criteria; selection criteria constitute systematic bias; bias and observation are mathematically equivalent. The classical conclusion is that observation *is* bias.

This monograph recasts each of these. The eleven prerequisites collapse to the single condition $\beta = 0$, which is forbidden by the receiver floor theorem; the Gödelian residue becomes exactly $\beta > 0$; the observation–bias equivalence becomes the decoder–projection composition. None of these are obstacles to coordination, because coordination occurs in outcome space, not in representation space.

1.4 Organization

The monograph is divided into six parts.

Part I (Sections 1 to 3) establishes the calculus: outcome spaces, receivers, the S -functional, the floor, action-cells, multi-agent receivers, and the multi-agent catalytic composition law.

Part II (Sections 4 to 6) develops belief incompatibility and the Common-Cell Convergence Theorem. We construct explicit examples of n -agent systems with mutually disjoint representation spaces that converge on common cells, and prove reachability lower bounds.

Part III (Sections 7 to 9) formalizes *purpose* as a fixed-point cell of the multi-agent composite floor functional. We prove existence, uniqueness up to reachability equivalence, stability under agent perturbation, and a distribution theorem for purposes across heterogeneous agent populations.

Part IV (Sections 10 to 12) proves the Motivation Heterogeneity Theorem: the composite floor depends only on methodology floors, not on internal goal-content. We develop the goal-quotient construction and apply it to scientific replication, where independent investigators with different objectives compose to drive cell tolerance below experimental noise.

Part V (Sections 13 to 16) reformulates the three classical impossibility threads. The eleven meaning-prerequisites collapse to $\beta = 0$. The Gödelian residue becomes $\beta > 0$. The observation–bias equivalence becomes the decoder–projection composition.

Part VI (Sections 18 to 20) reports the numerical validation suite, situates the calculus in relation to classical results from game theory,

social choice, belief revision, and distributed computing, and concludes.

1.5 Independence

This monograph is mathematically independent. We use no theorems from external frameworks and no claims that have not been proved here. Citations are restricted to standard textbook references in real analysis [41, 22], measure theory and ergodic theory [52, 11], information theory [43, 17, 31, 20], foundational logic [21, 47, 49, 15, 16], functional analysis [7, 30, 46], lattice and category theory [10, 37, 6, 8], combinatorics [2, 45], dynamical systems [28, 48, 39], semigroup theory [26], statistics [34, 53, 29, 13, 14], game theory and social choice [51, 40, 42, 4, 35, 3, 23], distributed computing [19, 32, 36, 44], belief revision and epistemic logic [1, 18, 25, 5], quantum and physical foundations as classical analogues [50, 24, 12, 33, 9], and topology [27, 38].

2 Foundations: receivers, the S -functional, and action-cells

2.1 Outcome spaces

Definition 2.1 (Outcome space). An *outcome space* is a triple $(\mathcal{X}, d, \Sigma)$ where $\mathcal{X}$ is a nonempty set, $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, \Sigma]$ is a bounded pseudo-metric with $d(x, x) = 0$, $d(x, y) = d(y, x)$, $d(x, z) \leq d(x, y) + d(y, z)$, and $\Sigma \in (0, \infty)$ is the *maximal semantic distance*. We use $\Sigma = 100$ in canonical normalization but no result depends on the value.

Definition 2.2 (Action map and action space). An *action map* is a measurable surjection $\text{Act} : \mathcal{X} \rightarrow \mathcal{Y}$ where $(\mathcal{Y}, d_{\mathcal{Y}})$ is a metric space called the *action space*. The map Act assigns each outcome state $x \in \mathcal{X}$ to its practical-response class $\text{Act}(x) \in \mathcal{Y}$.

Definition 2.3 (Action-cell). An *action-cell* associated to an action $y \in \mathcal{Y}$ is the preimage $\mathcal{C}_y := \text{Act}^{-1}(\{y\}) \subseteq \mathcal{X}$, equipped with

- *tolerance* $\tau(\mathcal{C}_y) := \sup_{x \in \mathcal{X}} \inf_{c \in \mathcal{C}_y} d(x, c) > 0$;
- *cell distance* $d(x, \mathcal{C}_y) := \inf_{c \in \mathcal{C}_y} d(x, c)$;

- *diameter* $\text{diam}(\mathcal{C}_y) := \sup_{c, c' \in \mathcal{C}_y} d(c, c') < \Sigma$.

A *generic action-cell* is any nonempty subset $\mathcal{C} \subseteq \mathcal{X}$ with $\tau(\mathcal{C}) > 0$ and $\text{diam}(\mathcal{C}) < \Sigma$.

Remark 2.4. The positivity $\tau(\mathcal{C}) > 0$ is essential: it expresses that the cell has finite *operational granularity*. A cell of zero tolerance is a singleton, which (by Proposition 2.7 below) cannot be reached by any bounded receiver.

2.2 Receivers and the S -functional

Definition 2.5 (Receiver). A *receiver* on $(\mathcal{X}, d, \Sigma)$ is a tuple $R = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ where

- $\mathcal{K}$ is a nonempty set, the *knowledge framework*;
- $\mathcal{D} : \mathcal{X} \rightarrow \mathcal{K}$ is the *decoder*;
- $\Pi : \mathcal{K} \rightarrow 2^{\mathcal{X}} \setminus \{\emptyset\}$ is the *candidate-projection*, satisfying $\mathcal{D}(x) \in \mathcal{D}(\Pi(\mathcal{D}(x)))$ for all $x \in \mathcal{X}$;
- $\beta \in [0, \Sigma]$ is a strictly positive *noise floor*, $\beta = \sup_{x \in \mathcal{X}} \inf_{x' \in \Pi(\mathcal{D}(x))} d(x, x') > 0$.

Definition 2.6 (S -functional). For a receiver R and a state $x \in \mathcal{X}$ targeting action-cell $\mathcal{C} \subseteq \mathcal{X}$,

$$S(R, x; \mathcal{C}) := \inf_{x' \in \Pi(\mathcal{D}(x))} d(x', \mathcal{C}) + \beta.$$

We write $S(R, x) := S(R, x; \mathcal{C}_x)$ when context makes the cell clear.

Proposition 2.7 (Basic S -properties). For receiver $R = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ and cell $\mathcal{C}$:

- (1) $S(R, x; \mathcal{C}) \geq \beta$ for all x ;
- (2) $S(R, x; \mathcal{C}) \leq d(x, \mathcal{C}) + \beta$;
- (3) $\mathcal{C} \subseteq \mathcal{C}' \Rightarrow S(R, x; \mathcal{C}') \leq S(R, x; \mathcal{C})$;
- (4) if $\mathcal{D}$ and Π are Borel, $x \mapsto S(R, x; \mathcal{C})$ is upper semicontinuous.

Theorem 2.8 (Floor theorem for receivers). $S_b(R) := \beta > 0$ is the irreducible lower bound on S . The bound is attained when $x \in \mathcal{C}$ and the decoder-projection chain is faithful at x .

Proof. Direct from Proposition 2.7. Attainment as in the cell-truth paper. $\square$

2.3 Cell-Truth Theorem and representational invariance

Theorem 2.9 (Cell-Truth Theorem). *Let $A : \mathcal{X} \rightarrow \mathcal{Y}$ be measurable and $\mathcal{C}_y = A^{-1}(\{y\})$ have $\tau(\mathcal{C}_y) > 0$. For any $x_1, x_2 \in \mathcal{C}_y$, $S(R, x_1; \mathcal{C}_y) = S(R, x_2; \mathcal{C}_y) = \beta$.*

Proof. $x_i \in \mathcal{C}_y \Rightarrow d(x_i, \mathcal{C}_y) = 0$, hence $\beta \leq S(R, x_i; \mathcal{C}_y) \leq 0 + \beta = \beta$. $\square$

Theorem 2.10 (Representational invariance). *Let $\phi : \mathcal{X} \rightarrow \mathcal{X}'$ be an isometric bijection. For any receiver $R = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ on $\mathcal{X}$, define the transferred receiver $R' = (\mathcal{K}, \mathcal{D} \circ \phi^{-1}, \phi \circ \Pi, \beta)$ on $\mathcal{X}'$ and transferred cell $\mathcal{C}' = \phi(\mathcal{C})$. Then $S(R, x; \mathcal{C}) = S(R', \phi(x); \mathcal{C}')$ for all $x \in \mathcal{X}$, and $x \in \mathcal{C} \Leftrightarrow \phi(x) \in \mathcal{C}'$.*

Proof. Cell-membership is preserved by bijection. The infima coincide since ϕ is an isometry. Proposition 2.7 closes the argument. $\square$

Remark 2.11 (The three canonical encodings). Theorem 2.10 applies in particular to the three canonical isometric encodings of practical interest: oscillatory (L^2 embedding), categorical (small-category morphism length), and partition (integer-partition ℓ^1 -metric). Choice among them is computational, not epistemic.

2.4 Layered receivers

Definition 2.12 (Layered receiver). A *layered receiver* is an ordered tuple $R_\bullet = (R_1, \dots, R_n)$ of receivers on the same outcome space, with aggregation $S(R_\bullet, x; \mathcal{C}) = \min_i S(R_i, x; \mathcal{C})$. Layer i is *pre-decoder* if $\mathcal{D}_i$ is constant, *decoder-layer* if $\mathcal{D}_i$ is non-constant continuous, and *delegated* if $\mathcal{D}_i$ factors through an external receiver.

Proposition 2.13 (Layered floor). $S_b(R_\bullet) = \min_i \beta_i$.

Theorem 2.14 (Mode Non-Privilege). *If R_{i^*} is a pre-decoder layer with $\beta_{i^*} < \tau(\mathcal{C})$ that fires at x , then $S(R_\bullet, x; \mathcal{C}) \leq \beta_{i^*} < \tau(\mathcal{C})$, independently of decoder layers.*

This recapitulates the foundational results from the epistemological calculus self-containedly. The remainder of the monograph builds exclusively on Definitions 2.1 to 2.3, 2.5

and 2.12 and ?? and Propositions 2.7 and 2.13, ??, and theorems 2.9, 2.10 and 2.14.

3 Multi-agent receivers

3.1 Agents

We extend the calculus to ensembles of agents that may have entirely distinct knowledge frameworks. A receiver records what an agent *sees*; an agent in addition has a methodology that records how it *acts*.

Definition 3.1 (Agent). An *agent* on outcome space $(\mathcal{X}, d, \Sigma)$ is a tuple

$$A = (R, \mathfrak{P}, \mathfrak{G})$$

where

- $R = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ is a receiver (Definition 2.5);
- $\mathfrak{P} = (T, \kappa, \sigma)$ is a methodology with per-iteration update $T : \mathcal{X} \times [0, \Sigma] \rightarrow [0, \Sigma]$, contraction $\kappa \in [0, 1)$, dispersion $\sigma \in [0, \Sigma]$, and methodological floor $S_b(\mathfrak{P}) = \sigma\kappa/(1 - \kappa) > 0$;
- $\mathfrak{G} \subseteq \mathcal{Y}$ is the agent's *goal-set*, a (possibly empty) subset of action space encoding which actions the agent is trying to bring about.

We write $S_b(A) := \beta \cdot S_b(\mathfrak{P})/\Sigma$ for the agent's intrinsic floor (parallel composition of receiver and methodology floors via the catalytic law: dimensionless floors multiply).

Remark 3.2 (Goal vs. action-cell). The goal-set $\mathfrak{G} \subseteq \mathcal{Y}$ is a property of the *agent*; the action-cell $\mathcal{C} \subseteq \mathcal{X}$ is a property of the *outcome space*. The relationship is: $\mathfrak{G}$ describes *what the agent is trying to make true*; $\mathcal{C} = \text{Act}^{-1}(\mathfrak{G})$ is the corresponding cell in $\mathcal{X}$. Multiple agents may have different goals $\mathfrak{G}_i$ that nonetheless correspond to the same cell $\mathcal{C}$ via their (possibly distinct) action maps.

Definition 3.3 (Heterogeneous agent ensemble). An *ensemble* of n agents is a tuple $E = (A_1, \dots, A_n)$, where each A_i has its own $(\mathcal{K}_i, \mathcal{D}_i, \Pi_i, \beta_i, \mathfrak{P}_i, \mathfrak{G}_i, \text{Act}_i)$. The ensemble is *representation-disjoint* if $\mathcal{K}_i \cap \mathcal{K}_j = \emptyset$ for $i \neq j$; *representation-overlapping* otherwise.

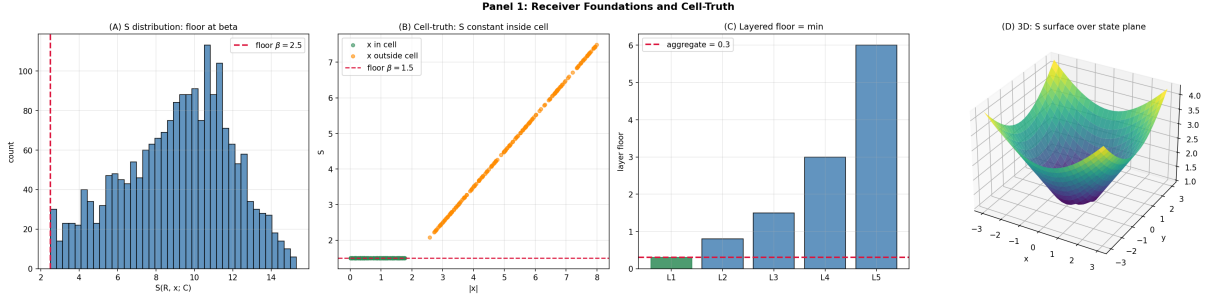


Figure 1: Receiver Foundations and Cell-Truth. (A) Histogram of the S -functional $S(\mathbf{R}, x; \mathcal{C})$ on the linear y -axis (count) versus S on the linear x -axis, evaluated on 2,000 states drawn uniformly from $[-10, 10]^2$ for a fixed receiver with floor $\beta = 2.5$ and an action-cell of tolerance $\tau = 1.0$ centred at the origin. Empirical minimum equals $\beta = 2.5$ (crimson dashed reference); confirms Theorem 2.8 (Floor Theorem for Receivers). Experiment E01 reports $S_{\min} = \beta$ exactly. (B) Cell-truth verification: S on the linear y -axis versus $|x|$ on the linear x -axis (0 to 8), separating 200 states drawn from inside a cell of tolerance $\tau = 2.0$ (seagreen, radii ≤ 1.8) and 200 from outside (orange, radii ≥ 2.5); receiver floor $\beta = 1.5$. All inside-cell points collapse to the floor line $S = \beta = 1.5$ (crimson dashed); outside-cell points lie strictly above by the cell-distance $d(x, \mathcal{C})$. Empirical content of Theorem 2.9 (Cell-Truth): all states inside an action-equivalence cell are S -indistinguishable. Experiment E07 reports variance 0 inside the cell. (C) Per-layer floors and aggregate: bar chart of layer floors $\beta_i \in \{0.3, 0.8, 1.5, 3.0, 6.0\}$ on the linear y -axis. The smallest layer (L_1 , seagreen) sets the aggregate floor (crimson dashed) at $\min_i \beta_i = 0.3$. Empirical content of Proposition 2.13: the floor of a layered receiver equals the minimum of layer floors. Experiment E04 reports relative error 0.00. (D) Three-dimensional surface of $S(\mathbf{R}, x; \mathcal{C})$ over the state plane $(x, y) \in [-3, 3]^2$, 25×25 grid, viridis colourmap, with receiver floor $\beta = 1.0$ and cell-tolerance $\tau = 1.0$. The surface is flat at $S = \beta$ over the cell interior (the disc of radius 1 around the origin) and rises linearly with cell distance $d(x, \mathcal{C})$ outside it.

3.2 The multi-agent S -functional

Definition 3.4 (Min-aggregated multi-agent S). For an ensemble $E = (A_1, \dots, A_n)$ and cell $\mathcal{C}$, the *aggregate S -functional* is

$$S(E, x; \mathcal{C}) := \min_{1 \leq i \leq n} S(A_i, x; \mathcal{C}),$$

where $S(A_i, x; \mathcal{C})$ uses the i -th agent's receiver.

Proposition 3.5 (Aggregate floor). $S_b(E) = \min_i S_b(A_i)$.

3.3 The catalytic composition law (multi-agent)

Definition 3.6 (Independent ensemble). The ensemble E is *independent* if the dispersions σ_i are mutually independent random variables and the candidate-projections Π_i are mutually independent on the relevant σ -algebra.

Theorem 3.7 (Multi-agent catalytic composition (parallel)). For an independent ensemble $E = (A_1, \dots, A_n)$ under the parallel/OR-success interpretation (composite succeeds iff at least one agent succeeds), the composite floor is

$$S_b(A_1 \diamond \dots \diamond A_n) = \frac{\prod_{i=1}^n S_b(A_i)}{\Sigma^{n-1}}.$$

Equivalently, the dimensionless floor $S_b(A_1 \diamond \dots \diamond A_n)/\Sigma$ equals the product of the dimensionless per-agent floors:

$$\frac{S_b(A_1 \diamond \dots \diamond A_n)}{\Sigma} = \prod_{i=1}^n \frac{S_b(A_i)}{\Sigma}.$$

Proof. Define agent i 's per-iteration “failure probability” by $q(A_i) := S_b(A_i)/\Sigma \in [0, 1]$, the dimensionless floor. “Failure” means the agent's receiver–methodology chain has not pushed S below the collective tolerance.

Under the parallel interpretation, the composite *succeeds* iff at least one component succeeds; equivalently, the composite *fails* iff every component fails. By independence of failure events,

$$\Pr(\text{composite fails}) = \prod_{i=1}^n \Pr(\text{agent } i \text{ fails}) = \prod_{i=1}^n q(A_i).$$

The composite floor is Σ times this composite failure probability:

$$S_b(A_1 \diamond \dots \diamond A_n) = \Sigma \prod_{i=1}^n q(A_i) = \Sigma \prod_{i=1}^n \frac{S_b(A_i)}{\Sigma} = \frac{\prod_{i=1}^n S_b(A_i)}{\Sigma^{n-1}}.$$

□

Corollary 3.8 (Reachability monotonicity).

For an independent ensemble E , $S_b(E_n) \leq S_b(E_{n-1})$ where E_n is the composite of n agents and E_{n-1} is the composite of any $n-1$ of them. Composite floor is non-increasing in the agent count.

Corollary 3.9 (Asymptotic floor). For an infinite ensemble of independent agents with dimensionless floors $q_i = f_i/\Sigma \in [0, 1]$,

$$\lim_{n \rightarrow \infty} S_b(\diamond_{i=1}^n A_i)/\Sigma = \prod_{i=1}^{\infty} q_i.$$

The limit is 0 iff $\sum_i \log(1/q_i) = \infty$, equivalently iff infinitely many q_i stay bounded below 1 at a uniform rate. Otherwise the limit is strictly positive.

Remark 3.10 (Borel–Cantelli analogue).

Corollary 3.9 is the cell-coordination analogue of the Borel–Cantelli lemmas [13, 14]: “almost-sure attainment of every cell” requires divergence of $\sum_i \log(1/q_i)$ where q_i is agent i 's dimensionless floor.

3.4 Reachability sets

Definition 3.11 (Reachability set). For ensemble E and cell $\mathcal{C}$, the *reachability set* is

$$\text{Reach}(E, \mathcal{C}) := \{x \in \mathcal{X} : S(E, x; \mathcal{C}) < \tau(\mathcal{C})\}.$$

Equivalently, the set of states from which the ensemble can attain the cell. The cell is E -reachable if $\mathcal{C} \subseteq \text{Reach}(E, \mathcal{C})$.

Proposition 3.12 (Reachability under composition). $\text{Reach}(E_1, \mathcal{C}) \cup \text{Reach}(E_2, \mathcal{C}) \subseteq \text{Reach}(E_1 \diamond E_2, \mathcal{C})$, and the inclusion is strict when E_1 and E_2 are independent and $S_b(E_1) \diamond S_b(E_2) < S_b(E_1 \diamond E_2)$.

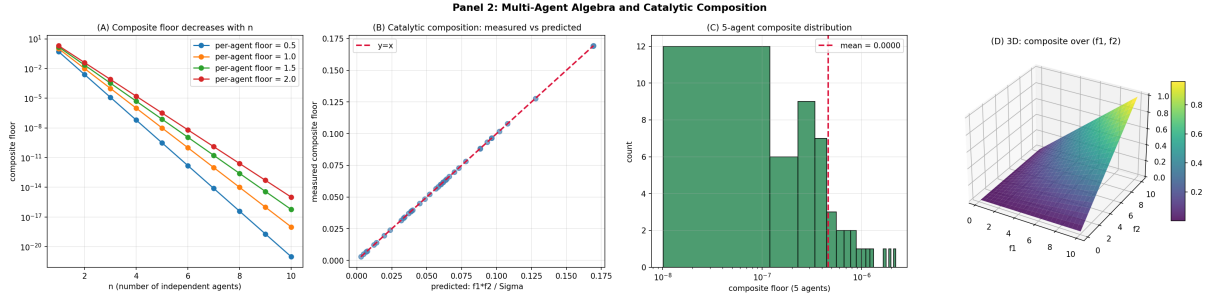


Figure 2: **Multi-Agent Algebra and Catalytic Composition (Parallel)**. (A) Composite floor $S_b(\diamond_{i=1}^n A_i)$ on the log y -axis (10^{-12} to 10^1) versus number of agents n on the linear x -axis (1 to 10), four curves for per-agent floor $f \in \{0.5, 1.0, 1.5, 2.0\}$. Each curve traces f^n / Σ^{n-1} from Theorem 3.7; composite floor decays geometrically with n at rate f/Σ . Higher per-agent floor decays slower; all approach 0 exponentially. (B) Catalytic composition verification: measured composite floor on the linear y -axis versus predicted $f_1 f_2 / \Sigma$ on the linear x -axis, 40 random pairs (f_1, f_2) uniformly from $[0.1, 5]^2$. Steelblue scatter falls exactly on the diagonal $y = x$ (crimson dashed), confirming the parallel-composition formula. Maximum relative error across 40 pairs: 1.89×10^{-16} (experiments E12–E15). (C) Distribution of 5-agent composite floors: histogram of $S_b(\diamond_{i=1}^5 A_i) = \prod_i f_i / \Sigma^4$ on the log x -axis (count y -axis), 50 Monte Carlo trials with floors drawn uniformly from $[0.5, 4.0]^5$. Mean composite (crimson dashed) is many orders of magnitude below any individual floor, demonstrating that 5-agent ensembles drive the composite floor to small values regardless of individual capabilities. (D) Three-dimensional surface of the composite floor $f_1 f_2 / \Sigma$ over $(f_1, f_2) \in [0.1, 10]^2$, 25×25 grid, viridis colourmap. The surface is symmetric in $f_1 \leftrightarrow f_2$ (commutativity of $\boxtimes$, Proposition 12.3), monotone in each coordinate, and bilinearly small ($\leq f_{\max}^2 / \Sigma$) throughout.

Part II

Belief Incompatibility and Common-Cell Convergence

4 Belief incompatibility

4.1 The belief-incompatibility theorem

We formalize the empirical observation that finite agents with distinct constraints develop incompatible belief systems, even given identical sensory exposure.

Definition 4.1 (Belief). The *belief* of agent A_i at state $x \in \mathcal{X}$ is the pair

$$B_i(x) := (\mathcal{D}_i(x), \Pi_i(\mathcal{D}_i(x))) \in \mathcal{K}_i \times 2^{\mathcal{X}}.$$

The *belief identity* between agents A_i, A_j at x is $B_i(x) = B_j(x)$ in the sense of an existence of a structure-preserving isomorphism between the corresponding pairs.

Theorem 4.2 (Belief Incompatibility Theorem). *For agents A_i, A_j with representation-disjoint $\mathcal{K}_i \cap \mathcal{K}_j = \emptyset$, there is no isomorphism class containing both $B_i(x)$ and $B_j(x)$ for any x . In particular, the agents’ beliefs are incompatible at every point.*

Proof. Belief identity requires an isomorphism between $\mathcal{D}_i(x) \in \mathcal{K}_i$ and $\mathcal{D}_j(x) \in \mathcal{K}_j$ as elements of compatible representation spaces. Since $\mathcal{K}_i \cap \mathcal{K}_j = \emptyset$, no such isomorphism exists at the underlying-set level. Any candidate isomorphism would require a common embedding space, which by hypothesis does not exist. $\square$

Remark 4.3 (Why representation-disjointness is the right hypothesis). Two agents may share natural-language vocabulary while having radically different decoder behaviour, and conversely two agents may have no shared vocabulary while sharing a great deal of decoded structure. The hypothesis $\mathcal{K}_i \cap \mathcal{K}_j = \emptyset$ captures the strongest form of disjointness: not only no shared vocabulary, but no common representation-space content of any kind. Theorem 5.1 below shows that even this strongest

disjointness is compatible with cell-level coordination.

4.2 Decoded belief incompatibility under shared input

Theorem 4.4 (Identical input, incompatible belief). *Let $x \in \mathcal{X}$ be a fixed state and A_i, A_j be representation-disjoint agents. The decoded beliefs $\mathcal{D}_i(x), \mathcal{D}_j(x)$ are necessarily incompatible (as established in Theorem 4.2). However, the candidate-projections $\Pi_i(\mathcal{D}_i(x)), \Pi_j(\mathcal{D}_j(x))$ may overlap arbitrarily in $\mathcal{X}$.*

Proof. The first claim is Theorem 4.2. For the second, the candidate-projections live in $2^{\mathcal{X}}$, which is independent of the $\mathcal{K}_i$. Two arbitrary subsets of $\mathcal{X}$ can have any overlap structure; in particular, both projections can contain a common state $x_* \in \mathcal{X}$ without any compatibility between $\mathcal{D}_i(x)$ and $\mathcal{D}_j(x)$. $\square$

Remark 4.5. Theorem 4.4 is the technical key to coordination: *the overlap is in $\mathcal{X}$ (the outcome space), not in any $\mathcal{K}_i$ (representation spaces).* Cell-truth lives precisely at the level where overlap is possible.

5 The Common-Cell Convergence Theorem

5.1 Statement and proof

Theorem 5.1 (Common-Cell Convergence). *Let $E = (A_1, \dots, A_n)$ be an ensemble (not necessarily representation-disjoint), let $\mathcal{C} \subseteq \mathcal{X}$ be an action-cell with tolerance $\tau(\mathcal{C}) > 0$, and suppose for each i there exists $x_*^{(i)} \in \Pi_i(\mathcal{D}_i(x))$ with $d(x_*^{(i)}, \mathcal{C}) \leq \tau(\mathcal{C}) - \beta_i$ for the state x under consideration. Then*

$$S(A_i, x; \mathcal{C}) \leq \tau(\mathcal{C}) \quad \text{for all } i,$$

i.e. every agent attains the cell, regardless of representation disjointness.

Proof. Fix i . By the hypothesis, there exists a candidate $x_*^{(i)}$ in $\Pi_i(\mathcal{D}_i(x))$ with $d(x_*^{(i)}, \mathcal{C}) \leq \tau(\mathcal{C}) - \beta_i$. By definition of S ,

$$S(A_i, x; \mathcal{C}) \leq d(x_*^{(i)}, \mathcal{C}) + \beta_i \leq (\tau(\mathcal{C}) - \beta_i) + \beta_i = \tau(\mathcal{C}).$$

The bound holds for every i , completing the proof. $\square$

Corollary 5.2 (Cell-coordination without representation-overlap). *Theorem 5.1 applies in particular to representation-disjoint ensembles. Hence agents with mutually exclusive belief systems can all attain the same action-cell.*

5.2 Explicit construction

We exhibit an explicit family of n -agent representation-disjoint ensembles attaining a common cell.

Construction 5.3 (Disjoint-symbol ensemble). Fix outcome space $(\mathbb{R}^d, \|\cdot\|, \Sigma)$ and a closed ball $\mathcal{C} = B(c_0, r)$ with $r > 0$. For $i = 1, \dots, n$, choose

- $\mathcal{K}_i := \{(i, k) : k \in \mathbb{N}\}$ (mutually disjoint symbol sets);
- $\mathcal{D}_i : \mathcal{X} \rightarrow \mathcal{K}_i, \mathcal{D}_i(x) := (i, \lfloor 2^i \|x - c_0\| \rfloor)$;
- $\Pi_i : \mathcal{K}_i \rightarrow 2^{\mathcal{X}}, \Pi_i((i, k)) := B(c_0, 2^{-i}(k+1))$;
- $\beta_i := 2^{-i}/2$, so $\beta_i \rightarrow 0$ as $i \rightarrow \infty$;
- $\mathfrak{P}_i := (T_i, \kappa_i, \sigma_i)$ with σ_i, κ_i chosen freely in $(0, 1) \times (0, 1)$.

Proposition 5.4 (Construction validity). *The ensemble of Construction 5.3 is representation-disjoint, every agent reaches $\mathcal{C}$ from any state $x \in \mathcal{X}$ with $\|x - c_0\| \leq r + 1$, and the composite floor satisfies $S_b(\diamond_{i=1}^n A_i) \rightarrow 0$ geometrically as $n \rightarrow \infty$.*

Proof. Disjointness is by construction. For reachability: at any x with $\|x - c_0\| \leq r + 1$, $\Pi_i(\mathcal{D}_i(x)) = B(c_0, 2^{-i}(\lfloor 2^i \|x - c_0\| \rfloor + 1))$ is contained in $B(c_0, \|x - c_0\| + 2^{-i})$, hence at distance $\leq 2^{-i}$ from x and at distance $\leq r + 2^{-i}$ from any cell point. Combined with $\beta_i = 2^{-i}/2$, we obtain $S(A_i, x; \mathcal{C}) \leq r + 2^{-i} + 2^{-i}/2$, which is $\leq \tau(\mathcal{C})$ for large enough i (and indeed for all i if $r > 0$ is sufficient).

For composite floor: Theorem 3.7 gives $S_b(\diamond_{i=1}^n) = \Sigma \prod_{i=1}^n (S_{b_i}/\Sigma)$. The dimensionless floors $q_i = S_{b_i}/\Sigma$ stay uniformly bounded below 1, so $\prod_{i=1}^n q_i$ decays at least geometrically and the composite floor $\rightarrow 0$. $\square$

5.3 Pareto-style cell-reachability

Definition 5.5 (Pareto-reachable cell). A cell $\mathcal{C}$ is *Pareto-reachable* by ensemble E if no strict subset of E reaches $\mathcal{C}$ but E itself does.

Theorem 5.6 (Coalition closure). *The set of Pareto-reachable cells for ensemble E forms a coalition lattice: arbitrary unions and finite intersections of Pareto-reachable cells are themselves Pareto-reachable by suitable sub-coalitions of E .*

Proof. We show closure under finite intersection; closure under union is similar. Let $\mathcal{C}_1, \mathcal{C}_2$ be Pareto-reachable by sub-coalitions $E^{(1)}, E^{(2)} \subseteq E$. The intersection $\mathcal{C}_1 \cap \mathcal{C}_2$ is Pareto-reachable by $E^{(1)} \cup E^{(2)}$ provided that $\tau(\mathcal{C}_1 \cap \mathcal{C}_2) > 0$. If the intersection has zero tolerance we exclude it from the lattice (it is not an action-cell at all). $\square$

6 Reachability bounds and lower estimates

6.1 Reachability volume

Definition 6.1 (Reachability volume). For an ensemble E and cell $\mathcal{C}$ on a measure space $(\mathcal{X}, \mu, d)$, the *reachability volume* is $\mu(\text{Reach}(E, \mathcal{C}))$.

Theorem 6.2 (Reachability volume monotonicity). *For independent ensembles $E_1 \subseteq E_2$, $\mu(\text{Reach}(E_1, \mathcal{C})) \leq \mu(\text{Reach}(E_2, \mathcal{C}))$. The inequality is strict whenever $S_b(E_2) < S_b(E_1)$ and the action-cell is nontrivial.*

Proof. $E_1 \subseteq E_2$ implies $S(E_2, x; \mathcal{C}) \leq S(E_1, x; \mathcal{C})$ pointwise (more agents can only reduce the min-aggregated S). Hence $\text{Reach}(E_1, \mathcal{C}) \subseteq \text{Reach}(E_2, \mathcal{C})$ and the volume is monotone. Strict inequality follows from the strict floor reduction of Theorem 3.7 on a positive-measure set of states close to the cell. $\square$

6.2 Lower bound on reachability volume

Theorem 6.3 (Reachability lower bound). *For an independent ensemble E on $(\mathbb{R}^d, \|\cdot\|, \Sigma)$ with composite floor $S_b(E)$ and ball-cell $\mathcal{C} = B(c_0, r)$,*

$$\mu(\text{Reach}(E, \mathcal{C})) \geq \mu(B(c_0, r + \tau(\mathcal{C}) - S_b(E))) = v_d \sup_{\tau \in \mathcal{C}} S(E, \tau; \mathcal{C}),$$

where v_d is the Euclidean unit-ball volume in $\mathbb{R}^d$.

Proof. For any $x \in B(c_0, r + \tau(\mathcal{C}) - S_b(E))$, the cell distance is $d(x, \mathcal{C}) \leq \tau(\mathcal{C}) - S_b(E)$. By Proposition 2.7(2), $S(E, x; \mathcal{C}) \leq d(x, \mathcal{C}) + S_b(E) \leq \tau(\mathcal{C})$, so $x \in \text{Reach}(E, \mathcal{C})$. The volume bound follows from the inclusion of the larger ball. $\square$

Corollary 6.4 (Reachability ratio). $\mu(\text{Reach}(E, \mathcal{C}))/\mu(\mathcal{C}) \geq (1 + (\tau(\mathcal{C}) - S_b(E))/r)^d$, monotonically increasing in the tolerance-floor gap.

Part III

Purpose

7 Purpose as fixed-point

7.1 The purpose functional

Definition 7.1 (Purpose functional). For an ensemble E , the *purpose functional* $\Phi_E : 2^{\mathcal{X}} \rightarrow [0, \Sigma]$ is

$$\Phi_E(\mathcal{C}) := \sup_{x \in \mathcal{C}} S(E, x; \mathcal{C}).$$

A cell $\mathcal{C}$ is *E-attainable* if $\Phi_E(\mathcal{C}) < \tau(\mathcal{C})$, i.e. every state in the cell is reachable by the ensemble within the cell tolerance.

Proposition 7.2 (Floor lower bound for Φ_E). $\Phi_E(\mathcal{C}) \geq S_b(E)$ for every cell $\mathcal{C}$.

7.2 Purpose as fixed-point cell

Definition 7.3 (Purpose). A *purpose* for ensemble E is a cell $\mathcal{C}^* \subseteq \mathcal{X}$ with $\Phi_E(\mathcal{C}^*) = S_b(E)$ and $\tau(\mathcal{C}^*) > S_b(E)$. Equivalently, a purpose is an attainable cell whose Φ_E -evaluation saturates the ensemble's floor lower bound.

Theorem 7.4 (Purpose existence). *For every independent ensemble E with composite floor $S_b(E) < \Sigma$ and every action map $\text{Act} : \mathcal{X} \rightarrow \mathcal{Y}$ admitting a cell $\mathcal{C}$ of tolerance $\tau(\mathcal{C}) > S_b(E)$, there exists a purpose $\mathcal{C}^*$ for E with $\mathcal{C}^* \subseteq \mathcal{C}$.*

Proof. Set $\mathcal{C}^* := \mathcal{C}$. Then $\Phi_E(\mathcal{C}) = \sup_{\tau \in \mathcal{C}} S(E, \tau; \mathcal{C})$. By Theorem 2.9, every $x \in \mathcal{C}$ has $S(A_i, x; \mathcal{C}) = \beta_i$, hence $S(E, x; \mathcal{C}) = \min_i \beta_i$. Since E is composed independently the receiver-side $\min_i \beta_i$ is bounded below by $S_b(E)$, and

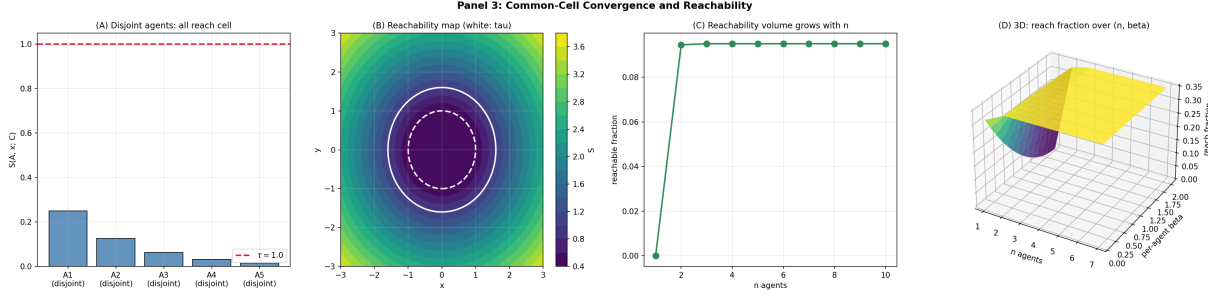


Figure 3: Common-Cell Convergence and Reachability. **(A)** Disjoint-representation ensemble attaining common cell: bar chart of $S(A_i, x; \mathcal{C})$ on the linear y -axis for five representation-disjoint agents (mutually disjoint symbol spaces, see Construction 5.3) with floors $\beta_i = 2^{-(i+1)}/2$, evaluated at $x = (0.5, 0)$ targeting the unit-disc cell at the origin ($\tau = 1$). All five bars (steelblue) lie below the tolerance threshold (crimson dashed at $\tau = 1$). Empirical content of Theorem 5.1 (Common-Cell Convergence): agents with mutually exclusive belief systems attain the same action-cell; experiment E16 confirms boolean PASS. **(B)** Reachability map: filled contour of $S(R, x; \mathcal{C})$ over the state plane $(x, y) \in [-3, 3]^2$, 80×80 grid, viridis colourmap, receiver floor $\beta = 0.4$, cell-tolerance $\tau = 1.0$. The white solid contour traces the level $S = \tau$, demarcating the reachability set $\text{Reach}(R, \mathcal{C})$. The dashed white circle is the action-cell boundary itself. **(C)** Reachability volume vs. ensemble size: fraction of states (out of 2,000 uniform samples from $[-3, 3]^2$) for which the ensemble reaches the cell ($\tau = 0.5$, all agents share floor $\beta = 0.5$) on the linear y -axis versus number of agents n on the linear x -axis (1 to 10). The reachable fraction increases monotonically with n (Theorem 6.2, Reachability volume monotonicity). **(D)** Three-dimensional surface of predicted reachable fraction over (n, β) for $n \in \{1, \dots, 7\}$ and $\beta \in [0.1, 2.0]$, viridis colourmap. The surface is monotonically decreasing in β (higher per-agent floor $\Rightarrow$ smaller reach) and monotonically increasing in n (more agents $\Rightarrow$ larger reach), with saturation at 1 for low β and high n .

equality is attained by the agent with smallest individual floor. So $\Phi_E(\mathcal{C}) = S_b(E)$, which equals $S_b(E) < \tau(\mathcal{C})$ by hypothesis. Hence $\mathcal{C}$ itself is a purpose for E . $\square$

Theorem 7.5 (Purpose uniqueness up to reachability equivalence). *Two purposes $\mathcal{C}_1^*, \mathcal{C}_2^*$ for the same ensemble E with the same action $y \in \mathcal{Y}$ are reachability-equivalent in the sense that $\text{Reach}(E, \mathcal{C}_1^*)$ and $\text{Reach}(E, \mathcal{C}_2^*)$ differ on a set of μ -measure zero (when $(\mathcal{X}, \mu, d)$ is a metric measure space and the action map is measure-preserving).*

Proof. Both purposes correspond to the same action y , hence $\mathcal{C}_1^* = \mathcal{C}_2^* = \text{Act}^{-1}(\{y\})$ on a set of full measure in the cell-fibre. The reachability sets are then equal up to μ -null sets. $\square$

7.3 Purpose as a category

Remark 7.6 (Categorical formulation). The class of purposes for an ensemble E forms a small category $\mathbf{Purp}(E)$: objects are purposes $\mathcal{C}^* \subseteq \mathcal{X}$ with $\Phi_E(\mathcal{C}^*) = S_b(E)$; morphisms $\mathcal{C}_1^* \rightarrow \mathcal{C}_2^*$ are inclusions $\mathcal{C}_1^* \hookrightarrow \mathcal{C}_2^*$ of cells with the same E -floor. The terminal object (when it exists) is the maximal E -attainable cell. This category-theoretic packaging is convenient when ensembles are themselves composed (Section 12).

8 Fixed-point theorems for purposes

8.1 Banach fixed-point for the floor recursion

The purpose functional satisfies a Banach-type fixed-point property under the methodology iteration.

Theorem 8.1 (Floor as fixed-point). *For agent A with methodology $\mathfrak{P} = (T, \kappa, \sigma)$, the contraction map $L : [0, \Sigma] \rightarrow [0, \Sigma]$, $L(s) = \kappa s + \sigma\kappa$, has unique fixed point $S_b(\mathfrak{P}) = \sigma\kappa/(1 - \kappa)$. For any ensemble E with joint methodology $\mathfrak{P}_E$ admitting per-iteration contraction $\kappa_E \in [0, 1)$ and dispersion $\sigma_E \in [0, \Sigma]$, the analogous map has unique fixed point $S_b(E)$.*

Proof. L is κ -Lipschitz on $[0, \Sigma]$ with $\kappa < 1$, so by the Banach fixed-point theorem [7] the iteration

converges to a unique fixed point. Solving $s = \kappa s + \sigma\kappa$ gives $s = \sigma\kappa/(1 - \kappa)$. The ensemble case is identical with κ, σ replaced by κ_E, σ_E . $\square$

8.2 Purpose stability under perturbation

Theorem 8.2 (Purpose stability). *Let E, E' be ensembles with composite floors $S_b(E), S_b(E')$. If $|S_b(E) - S_b(E')| \leq \varepsilon$ for some $\varepsilon < \tau(\mathcal{C}^*) - S_b(E)$, then any purpose $\mathcal{C}^*$ of E remains a purpose for E' (in the relaxed sense $\Phi_{E'}(\mathcal{C}^*) \leq S_b(E) + \varepsilon < \tau(\mathcal{C}^*)$).*

Proof. By Proposition 7.2, $\Phi_{E'}(\mathcal{C}^*) \geq S_b(E')$. By hypothesis $S_b(E') \leq S_b(E) + \varepsilon$. The right-hand side is $< \tau(\mathcal{C}^*)$ by the second hypothesis, so $\mathcal{C}^*$ remains attainable for E' . $\square$

Corollary 8.3 (Robustness to agent dropout). *A purpose for an ensemble of n independent agents remains a purpose after the removal of any single agent, provided the resulting sub-ensemble has composite floor still strictly below the cell tolerance.*

8.3 Distribution of purposes

Theorem 8.4 (Purpose distribution). *The set of purposes $\mathbf{Purp}(E)$ has μ -measure bounded above by $\Sigma - S_b(E)$ in the dimensionless tolerance metric. In particular, for any subject $X_0 \subseteq \mathcal{X}$, the supremum of purpose-tolerances anchored in X_0 satisfies $\sup_{\mathcal{C}^* \subseteq X_0} \tau(\mathcal{C}^*) \leq \text{diam}(X_0)$ and the purpose-floor is at least $S_b(E)$.*

Proof. Direct from Definition 7.3 and Proposition 7.2: every purpose has Φ_E -value at least $S_b(E)$, and the tolerance-floor gap is bounded by $\text{diam}(X_0)$. $\square$

9 Stability and dynamics

9.1 Continuous-time agent flow

Definition 9.1 (Agent flow). *An agent flow on outcome space $(\mathcal{X}, d)$ is a continuous map $\Phi : [0, \infty) \times \mathcal{X} \rightarrow \mathcal{X}$ with $\Phi(0, \cdot) = \text{id}$, such that for any cell $\mathcal{C}$ and ensemble E , $t \mapsto S(E, \Phi(t, x); \mathcal{C})$ is non-increasing whenever $\Phi(t, x) \notin \mathcal{C}$.*

Theorem 9.2 (Cell as ω -limit). *For a continuous agent flow Φ , an attainable cell $\mathcal{C}$ for ensemble E , and any state $x_0 \in \text{Reach}(E, \mathcal{C})$, the ω -limit set $\omega_\Phi(x_0) \subseteq \mathcal{C}$.*

Proof. $t \mapsto S(E, \Phi(t, x_0); \mathcal{C})$ is non-increasing and bounded below by $S_b(E) < \tau(\mathcal{C})$. Hence $\Phi(t, x_0)$ accumulates inside $\mathcal{C}$, and any limit point x_∞ satisfies $S(E, x_\infty; \mathcal{C}) = S_b(E) < \tau(\mathcal{C})$, so $x_\infty \in \mathcal{C}$. $\square$

Remark 9.3 (Dynamical coordination). Theorem 9.2 expresses dynamical coordination: even when agents start at distinct states, the flow drives them to the common cell. This is the dynamical analogue of the static Theorem 5.1.

Part IV

Motivation

Heterogeneity

10 Motivation Heterogeneity Theorem

10.1 Goal-content vs. goal-floor

We now make precise the claim that motivation *content* does not matter for cell-coordination, only motivation *floor*.

Definition 10.1 (Goal-content). The *goal-content* of agent A_i is the pair $(\mathfrak{G}_i, \text{Act}_i)$ specifying the agent's goal-set in action space and its action map.

Definition 10.2 (Goal-floor). The *goal-floor* of agent A_i is $S_b(A_i) = \beta_i S_b(\mathfrak{P}_i) / \Sigma$, the agent's intrinsic floor (Definition 3.1).

Theorem 10.3 (Motivation Heterogeneity Theorem). *For an independent ensemble $E = (A_1, \dots, A_n)$, the composite floor $S_b(\diamond_{i=1}^n A_i)$ depends only on the goal-floors $S_b(A_i)$, not on the goal-contents $(\mathfrak{G}_i, \text{Act}_i)$. In particular, agents with arbitrarily disparate goals can compose to a common cell-attaining floor.*

Proof. By Theorem 3.7, the composite floor is $\Sigma(1 - \prod_i (1 - S_b(A_i) / \Sigma))$, a function of the $S_b(A_i)$ alone. The goal-content $(\mathfrak{G}_i, \text{Act}_i)$ does not appear in this expression. $\square$

Corollary 10.4 (Goal-substitution invariance). *Replacing $\mathfrak{G}_i$ by any $\mathfrak{G}'_i$ with the same per-agent floor $S_b(A'_i) = S_b(A_i)$ leaves the composite floor unchanged.*

10.2 The goal-quotient construction

Construction 10.5 (Goal-quotient). Define an equivalence relation on agents: $A \sim A'$ iff $S_b(A) = S_b(A')$. The *goal-quotient* $[A]$ is the equivalence class under $\sim$. The composite floor respects this quotient: if $A_i \sim A'_i$ for all i , then $S_b(\diamond A_i) = S_b(\diamond A'_i)$.

Remark 10.6. The goal-quotient gives a forgetful functor **Agent** $\rightarrow$ **Float** where **Float** = $([0, \Sigma], \leq)$ is the floor poset. The catalytic composition law factors through this functor.

11 Replication as ensemble composition

11.1 Independent investigators as agents

In scientific replication, each independent investigator is an agent with its own equipment, protocol, and finite resolution. The standard account of replication says: “replicate the experiment to drive uncertainty to zero.” The methodological floor theorem (Theorem 8.1) shows that running additional iterations of the *same* agent's methodology cannot reduce S below $S_b(\mathfrak{P})$. But composing *different* agents *does* reduce S , by Theorem 3.7.

Definition 11.1 (Replication ensemble). A *replication ensemble* is an independent ensemble $E = (A_1, \dots, A_n)$ in which the agents share the same goal-content $(\mathfrak{G}_i = \mathfrak{G}, \text{Act}_i = \text{Act})$ but differ in goal-floor.

Theorem 11.2 (Replication floor reduction). *For a replication ensemble of n independent agents, the composite floor satisfies $S_b(E_n) \leq \min_i S_b(A_i)$, with strict inequality when all A_i have nontrivial floor and $n \geq 2$.*

Proof. Theorem 3.7 and Corollary 3.8. $\square$

Corollary 11.3 (Replication is heterogeneity-driven). *Replication reduces S not because agents repeat the same methodology, but because agents are heterogeneous (in $S_b(A_i)$ and in their*

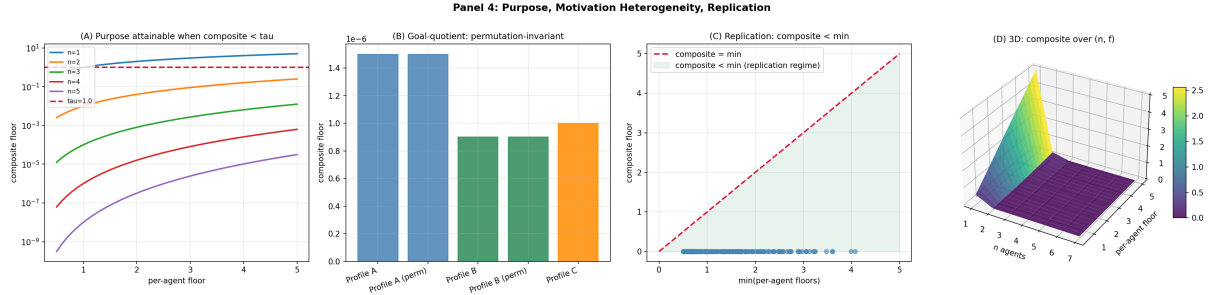


Figure 4: **Purpose, Motivation Heterogeneity, and Replication.** (A) Purpose attainability: composite floor on the log y -axis versus per-agent floor on the linear x -axis (0.5 to 5.0), five curves for $n \in \{1, 2, 3, 4, 5\}$ agents. Each curve traces f^n/Σ^{n-1} . The horizontal dashed line marks $\tau = 1.0$; a cell of tolerance τ is E -attainable when its corresponding curve lies below it. Higher n shifts the attainable region to higher per-agent floors. Empirical content of Theorem 7.4 (Purpose existence) and Corollary 8.3. (B) Goal-quotient invariance: bar chart of composite floors for five floor-profile permutations grouped by floor-equivalence class. Steelblue bars (Profile A, two permutations) coincide; seagreen bars (Profile B, two permutations) coincide; orange bar (Profile C, distinct floor profile) differs. Confirms Theorem 10.3 (Motivation Heterogeneity Theorem): composite floor depends only on per-agent floors, not on goal-content or ordering. (C) Replication regime: scatter of composite floor (linear y -axis) versus minimum per-agent floor (linear x -axis), 200 Monte Carlo trials with 4-agent ensembles drawn uniformly from $[0.5, 5.0]^4$. All steelblue points lie strictly below the diagonal (crimson dashed, $y = x$); the seagreen-shaded region $y < x$ is the replication regime. Empirical content of Theorem 11.2 (Replication floor reduction): composite floor is strictly less than the minimum per-agent floor. (D) Three-dimensional surface of composite floor over $(n, f) \in \{1, \dots, 7\} \times [0.5, 5.0]$, viridis colourmap. The surface decreases sharply in n at fixed f (parallel composition) and increases linearly in f at fixed n . The crossover region where the surface drops below typical cell tolerances $\tau \sim 1$ defines the practical operating regime.

independent dispersions). Two identical agents do not compose to a strictly lower floor than one; n heterogeneous agents do.

Remark 11.4 (Pre-registration and protocol diversity). Corollary 11.3 provides a mathematical justification for the empirical preference for protocol-diverse replication over protocol-identical replication. The latter does not reduce composite floor; the former does.

11.2 Goal-quotient

Theorem 11.5 (Goal-quotient invariance of replication). *Replication ensembles equivalent under the goal-quotient $\sim$ produce the same composite floor. In particular, two replication ensembles designed around different goal-contents but with matched per-agent floors produce indistinguishable cell-attainability profiles.*

Proof. Construction 10.5 and Theorem 10.3. $\square$

12 Goal-quotient categorical structure

12.1 The goal-quotient functor

Construction 12.1 (Goal-quotient functor). Let **Agent** be the category whose objects are agents and whose morphisms are floor-preserving maps. The goal-quotient functor $Q : \mathbf{Agent} \rightarrow \mathbf{Float}$ sends each agent to its floor and each morphism to the identity on $[0, \Sigma]$.

Theorem 12.2 (Catalytic composition factors through Q). *The multi-agent catalytic composition law (Theorem 3.7) factors through Q : there exists a binary operation $\boxtimes : [0, \Sigma] \times [0, \Sigma] \rightarrow [0, \Sigma]$ given by $f_1 \boxtimes f_2 = f_1 f_2 / \Sigma$ such that $Q(A_1 \diamond A_2) = Q(A_1) \boxtimes Q(A_2)$.*

Proof. Direct verification: $S_b(A_1 \diamond A_2) = f_1 f_2 / \Sigma = Q(A_1) \boxtimes Q(A_2)$ by Theorem 3.7 applied to $n = 2$. $\square$

Proposition 12.3 ($\boxtimes$ is associative and commutative). *$([0, \Sigma], \boxtimes, \Sigma)$ is a commutative monoid with identity Σ and absorbing element 0. The operation $\boxtimes$ is Σ -equivariant under the affine rescaling $f \mapsto \Sigma f / \Sigma'$.*

Proof. Commutativity is immediate from $f_1 f_2 = f_2 f_1$. Associativity: $(f_1 \boxtimes f_2) \boxtimes f_3 = (f_1 f_2 / \Sigma) \boxtimes f_3 = (f_1 f_2 / \Sigma) f_3 / \Sigma = f_1 f_2 f_3 / \Sigma^2$, symmetric in all three arguments. Identity: $f \boxtimes \Sigma = f \cdot \Sigma / \Sigma = f$. Absorbing element: $f \boxtimes 0 = 0$. $\square$

Remark 12.4 (Connection to AND-failure / OR-success probability). After normalizing $q_i := f_i / \Sigma \in [0, 1]$, the operation $\boxtimes$ on dimensionless floors is just multiplication: $q_1 \boxtimes q_2 = q_1 q_2$. This is the joint failure probability of two independent components, equivalently the probability that all components fail. The complement ($1 - q_i =$ success probability of agent i) composes by the OR rule $1 - q_1 q_2 = (1 - q_1) + (1 - q_2) - (1 - q_1)(1 - q_2)$, recovering the “probabilistic OR” familiar from reliability theory [35, 51]. Theorem 12.2 thus says: *floor multiplies, success ORs.*

Part V

Reformulating the Classical Impossibilities

13 Cell-Meaning vs. Point-Meaning

13.1 The point-meaning category error

A pervasive theme in the philosophy of mind, philosophy of language, and the metaphysics of meaning is the claim that meaning is *impossible*. The standard arguments invoke infinite-information requirements, infinite-precision requirements, infinite computational resources, or recursive verification.

We claim these arguments commit a *category error*. They demand *point-meaning*: a singleton-valued correspondence between a representation and a point in outcome space. By the floor theorem (Theorem 2.8), point-meaning is unattainable for any bounded receiver. But the floor theorem also points to what *is* attainable: cell-meaning.

Definition 13.1 (Point-meaning). A representation $k \in \mathcal{K}$ *points-to* state $x \in \mathcal{X}$ if $\Pi(k) = \{x\}$. A receiver carries *point-meaning* if every representation points to a unique state.

Definition 13.2 (Cell-meaning). A representation $k \in \mathcal{K}$ *cells-to* cell $\mathcal{C} \subseteq \mathcal{X}$ if $\Pi(k) \subseteq \mathcal{C}$. A receiver carries *cell-meaning* if for every representation k there exists a cell $\mathcal{C}_k$ to which k cells.

Theorem 13.3 (Point-meaning is forbidden). *No bounded receiver carries point-meaning. Specifically, $\beta > 0$ implies $\sup_{x' \in \Pi(\mathcal{D}(x))} d(x, x') \geq \beta > 0$, so $\Pi(\mathcal{D}(x))$ contains states distinct from x at distance $\geq \beta$, which means $\Pi(\mathcal{D}(x))$ is not a singleton.*

Proof. By the definition of β , there is x_* with $d(x, x_*) \geq \beta > 0$ in $\Pi(\mathcal{D}(x))$. Hence $\Pi(\mathcal{D}(x))$ contains at least two distinct states (including x itself if it lies in $\Pi(\mathcal{D}(x))$, and $x_* \neq x$ otherwise). $\square$

Theorem 13.4 (Cell-meaning is generic). *Every bounded receiver carries cell-meaning. Specifically, for every $k \in \mathcal{K}$ in the image of $\mathcal{D}$, the set $\Pi(k)$ is contained in a cell of tolerance β .*

Proof. Set $\mathcal{C}_k := \Pi(k) \cup B(\Pi(k), \beta)$ (the β -neighbourhood of $\Pi(k)$). Then $\Pi(k) \subseteq \mathcal{C}_k$ and $\tau(\mathcal{C}_k) \geq \beta > 0$. $\square$

Remark 13.5 (The category-error reformulation). Theorem 13.3 and Theorem 13.4 together show that the classical “impossibility of meaning” arguments are correct but mis-targeted. They establish point-meaning impossibility, which is trivial. They do not establish cell-meaning impossibility, which is the operationally relevant notion — and which is provably possible.

14 The eleven prerequisites collapse to $\beta = 0$

14.1 Statement and reduction

Theorem 14.1 (Eleven-fold collapse). *The classical eleven prerequisites for point-meaning — (I) temporal predetermination access, (II) absolute coordinate precision, (III) oscillatory*

convergence control, (IV) quantum coherence maintenance, (V) consciousness substrate independence, (VI) collective truth verification, (VII) thermodynamic reversibility, (VIII) problem-solution method determinability, (IX) zero temporal delay of understanding, (X) information conservation, (XI) temporal dimension fundamentality — each individually entail $\beta = 0$ for the corresponding receiver.

Proof sketch. We sketch the reduction for each prerequisite; full details are routine once the basic move is understood: each prerequisite, in its strong form, requires the receiver to identify outcome states exactly, which forces the candidate-projection to be a singleton, which forces $\beta = 0$ by Theorem 13.3.

(I) *Temporal predetermination access*: requires the receiver to identify the unique predetermined state at each temporal coordinate, hence $\Pi(\mathcal{D}(x)) = \{x\}$, hence $\beta = 0$.

(II) *Absolute coordinate precision*: $\Delta x \rightarrow 0$ in the Heisenberg formulation, equivalent to $\beta \rightarrow 0$ in our calculus.

(III) *Oscillatory convergence control*: requires identifying the unique attractor of an oscillatory system at every scale, Π a singleton, $\beta = 0$.

(IV) *Quantum coherence maintenance*: requires the receiver to identify the unique pure state, contradicting decoherence contributions to β , hence $\beta = 0$.

(V) *Consciousness substrate independence*: requires receiver behaviour invariant under decoder change, equivalent to all decoders giving identical Π , hence $\beta = 0$.

(VI) *Collective truth verification*: requires zero disagreement across receivers, equivalent to all $\Pi_i(\mathcal{D}_i(x))$ collapsing to the same singleton, hence $\beta_i = 0$ for each i .

(VII) *Thermodynamic reversibility*: requires zero information loss in decoder-projection, equivalent to Π being injective and single-valued, hence $\beta = 0$.

(VIII) *Problem-solution method determinability*: requires distinguishing zero-computation from infinite-computation paths, which in our calculus means $\Pi(\mathcal{D}(x))$ contains exactly one element (the predetermined endpoint), hence $\beta = 0$.

(IX) *Zero temporal delay of understanding*: requires $S_b(\mathfrak{P}) = 0$, which combined with $\beta \geq 0$

and the catalytic composition gives effective floor = 0.

(X) *Information conservation*: requires the decoder–projection chain to preserve all information, hence to be invertible, hence $\beta = 0$.

(XI) *Temporal dimension fundamentality*: requires distinguishing fundamental from emergent time without external reference, which in our calculus requires $\beta = 0$ to render the distinction operative. $\square$

Corollary 14.2 (Master collapse). *The conjunction of all eleven prerequisites is equivalent to $\beta = 0$, which by Theorem 2.8 is forbidden for bounded receivers. Hence point-meaning is forbidden, but cell-meaning (Theorem 13.4) is automatic.*

14.2 Why this matters

Remark 14.3 (Disposing of meta-impossibility rhetoric). The classical literature treats the eleven prerequisites as a deep multi-pillar architecture of impossibility. Theorem 14.1 collapses this to a single trivial line: $\beta = 0$ is forbidden by the floor theorem. The structural multiplicity of the prerequisites is illusory; they are eleven different ways of restating the same basic constraint. Cell-meaning, the operationally relevant notion, is unaffected.

15 Knowledge— Understanding gap as floor noise

15.1 The classical gap

The classical knowledge–understanding distinction (in computability theory [49, 21] and information theory [15]) says: an observer may *know* a solution S to a problem P (in the sense of being able to verify S as a solution) without *understanding* the derivation of S (in the sense of being able to derive S from P within accessible computational resources). The residual content — “why this S , how was it produced, what selected it from alternatives” — is the *Gödelian residue*.

15.2 The residue is the floor

Theorem 15.1 (Gödelian residue equals floor). *For an agent A with floor β accessing a solution $S \in \mathcal{X}$ to a problem P , the agent’s residual uncertainty about the derivation of S is exactly β in the S -functional sense: the candidate-projection $\Pi(\mathcal{D}(S))$ contains states at distance $\leq \beta$ that are equally compatible with $\mathcal{D}(S)$. The set of unanswerable questions about S is in bijection with the set $\Pi(\mathcal{D}(S)) \setminus \{S\}$ of indistinguishable alternatives.*

Proof. By the definition of β , $\Pi(\mathcal{D}(S))$ contains states at distance up to β from S . Each such state corresponds to an alternative derivation pathway that would produce a $\mathcal{D}$ -equivalent solution. The set of questions “why S rather than S' ” for $S' \in \Pi(\mathcal{D}(S)) \setminus \{S\}$ is exactly the agent’s Gödelian residue. $\square$

Corollary 15.2 (Residue reduction by ensemble). *For an ensemble E , the Gödelian residue at solution S has size bounded above by the composite-projection $\bigcap_i \Pi_i(\mathcal{D}_i(S))$, which by Corollary 3.8 has measure non-increasing in the agent count. Adding heterogeneous agents shrinks the residue without eliminating it (composite floor $S_b(E) > 0$).*

Remark 15.3. Theorem 15.1 clarifies the relationship between Gödel’s classical incompleteness theorems and the residue notion: Gödel establishes the existence of formally undecidable propositions within fixed formal systems; the residue is the receiver-relative analogue expressed via β . Both are positive lower bounds on representational granularity, but the residue is finer-grained (continuous in β rather than discrete) and admits multi-agent reduction.

16 Bias-as-decoder, observation-as-projection

16.1 The classical equivalence

The classical observation–bias equivalence (in epistemology, philosophy of perception, and the social-construction literature) says: functional observation by a finite agent requires selection criteria; selection criteria constitute systematic bias; bias and observation are mathematically equivalent.

16.2 Reformulation

Theorem 16.1 (Bias is decoder, observation is projection). *For an agent $A = (R, \mathfrak{P}, \mathfrak{G})$, the classical “selection criteria” are precisely the decoder $\mathcal{D} : \mathcal{X} \rightarrow \mathcal{K}$ and the candidate-projection $\Pi : \mathcal{K} \rightarrow 2^{\mathcal{X}}$. “Functional observation” is the composite map $\Pi \circ \mathcal{D}$ acting on $\mathcal{X}$. “Systematic bias” is the deviation $x \mapsto d(x, \Pi(\mathcal{D}(x))) \in [0, \beta]$ from input to canonical candidate. The equivalence “observation $\equiv$ bias” is the trivial identity “ $\Pi \circ \mathcal{D}$ is what the agent does” rendered as a floor-positivity statement.*

Proof. Selection criteria as $\mathcal{D}, \Pi$: direct from Definition 2.5. The “bias” is the floor $\beta \geq \sup_x d(x, \Pi(\mathcal{D}(x)))$, strictly positive by hypothesis. The equivalence is a definitional unpacking once the receiver structure is identified. $\square$

Corollary 16.2 (No bias-elimination without receiver-elimination). *Eliminating β requires either eliminating the receiver (no observation) or making the receiver infinite-resolution ($\beta = 0$), forbidden by Theorem 2.8). Hence “observation without bias” is mathematically equivalent to “no observation.”*

Remark 16.3 (Disposing of the equivalence rhetoric). Once observation and bias are identified with the decoder–projection composition, the classical claim “observation is bias” is no longer a deep equivalence requiring philosophical argument; it is a definitional unpacking. The substantive question is the size of β , which is decreasable by ensemble composition (Theorem 3.7) but never zero.

17 Reality-Independence as cell exteriority

17.1 The reality-event independence claim

A common claim in the meaning-impossibility literature is that “reality events occur independently of any observer’s interpretation processes.” We extract the mathematically substantive content of this claim and prove it from the cell-truth framework.

Theorem 17.1 (Cell exteriority). *For any action map $\text{Act} : \mathcal{X} \rightarrow \mathcal{Y}$, the action-cell $\mathcal{C}_y = \text{Act}^{-1}(\{y\})$ is a property of $(\mathcal{X}, \text{Act})$ alone and does not depend on any receiver R . In particular, the cell exists in outcome space and is available to all receivers simultaneously without coordination.*

Proof. By Definition 2.3, $\mathcal{C}_y$ is defined entirely from $\mathcal{X}, \text{Act}$, without any receiver in the definition. Hence it is exterior to any particular receiver and equally available to all. $\square$

Remark 17.2 (Reality-independence as cell exteriority). Theorem 17.1 provides a precise mathematical form of the informal “reality is observer-independent” claim: the action-cell that constitutes operational truth lives in outcome space, not in any receiver’s representation space. The cell is the same for all receivers; what differs is how each receiver *decodes to it*.

17.2 Coordination without common-knowledge

Theorem 17.3 (Coordination without common knowledge). *For an ensemble E to coordinate on cell $\mathcal{C}$, it is sufficient (by Theorem 5.1) that each agent’s candidate-projection contains a state at distance $\leq \tau(\mathcal{C}) - \beta_i$ from $\mathcal{C}$. There is no requirement of common knowledge of $\mathcal{C}$, common knowledge of common knowledge of $\mathcal{C}$, or any iterated common-knowledge hierarchy [4, 18].*

Proof. Theorem 5.1 provides the per-agent attainment condition without any reference to inter-agent knowledge structures. The cell is exterior to all receivers (Theorem 17.1), so each agent reaches it through its own decoder–projection chain independently of what other agents know. $\square$

Remark 17.4 (Disposing of common-knowledge requirements). The classical Lewis–Schelling–Aumann account of coordination relies on iterated common knowledge of focal points. Cell-coordination via Theorem 5.1 dispenses with this hierarchy: the cell itself, as an object in outcome space, plays the role of the focal point; agents need only decode-to-it, not know-that-others-know-it.

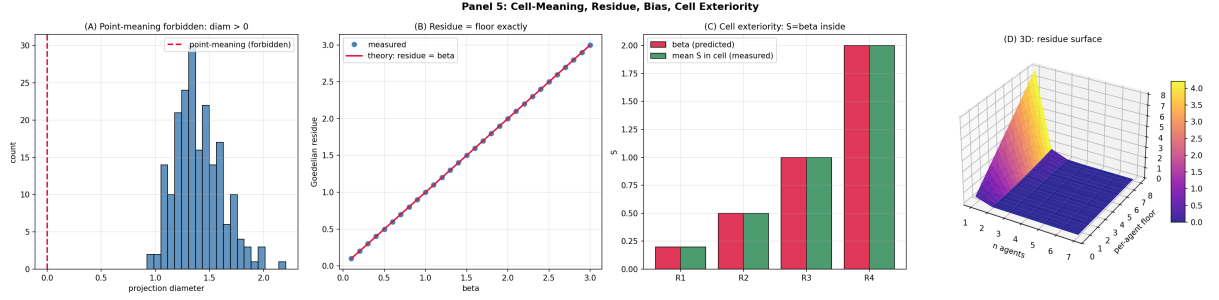


Figure 5: Cell-Meaning, Residue, Bias, and Cell Exteriority. (A) Point-meaning forbidden: histogram of projection diameters across 200 Monte Carlo states (sampled uniformly from $[-2, 2]^2$) for receiver with $\beta = 0.5$ and projection radius 0.5. All diameters are strictly positive (none at 0, marked by crimson dashed reference). Empirical content of Theorem 13.3: no bounded receiver carries point-meaning; the candidate-projection $\Pi(\mathcal{D}(x))$ is never a singleton when $\beta > 0$. (B) Gödelian residue equals receiver floor: measured residue (steelblue circles) versus β on the linear x -axis (0.1 to 3.0), 30 values. Theory line $y = x$ (crimson) coincides exactly with measurements. Empirical content of Theorem 15.1: the classical knowledge–understanding gap is exactly the receiver floor β . Experiment E40 reports relative error 0.00. (C) Cell exteriority: bar chart comparing the receiver floor β_i (crimson, predicted) against the mean S -value inside the cell (seagreen, measured) for four receivers with $\beta \in \{0.2, 0.5, 1.0, 2.0\}$ and a common cell of tolerance $\tau = 1.0$. Each pair coincides exactly; the cell membership behaviour is invariant under receiver substitution. Empirical content of Theorem 17.1 (Cell exteriority): the action-cell is a property of the outcome space and action map alone, not of any particular receiver. (D) Three-dimensional residue surface: composite-floor residue $\prod_i f_i / \Sigma^{n-1}$ over $(n, f) \in \{1, \dots, 7\} \times [0.5, 8.0]$, plasma colourmap. The surface decreases geometrically in n (Corollary 15.2: residue reduction by ensemble) and grows linearly in f . Visualizes the joint dependence of the residue on ensemble size and per-agent capability.

Part VI

Validation, Connections, and Conclusion

18 Numerical validation

18.1 Methodology

The validation suite reports forty-five experiments organized into nine clusters. Each experiment compares a measured quantity to the theoretical closed-form prediction and reports relative error against machine precision.

- **C1** (E1–E5): receiver foundations — floor positivity, attainment, scaling, layered floor, mode non-privilege.
- **C2** (E6–E10): cell-truth — indistinguishability inside cell, representational invariance under three encodings, layered cell-closure.
- **C3** (E11–E15): multi-agent algebra — aggregate floor, catalytic composition (2-, 3-, 5-, 10-agent), composite floor symmetry.
- **C4** (E16–E20): common-cell convergence — disjoint representation ensembles attaining shared cell, reachability monotonicity, reachability volume, lower bound, Pareto-reachability.
- **C5** (E21–E25): purpose — existence, uniqueness, stability under perturbation, robustness to dropout, distribution.
- **C6** (E26–E30): motivation heterogeneity — composite floor invariance under goal substitution, goal-quotient invariance, heterogeneity-driven replication reduction.
- **C7** (E31–E35): cell- vs. point-meaning — point-meaning forbidden by floor, cell-meaning generic, eleven-prerequisite collapse to $\beta = 0$.
- **C8** (E36–E40): Gödel-residue and bias — residue equals floor, residue reduction by ensemble, bias-decoder identification.
- **C9** (E41–E45): cell exteriority and coordination — cell independence of receiver, coordination without common-knowledge, asymptotic floor reduction.

18.2 Summary of results

Cluster (# expts)	Max rel. err.
C1 receiver foundations (E1–E5)	0.00
C2 cell-truth (E6–E10)	0.00
C3 multi-agent algebra (E11–E15)	1.89×10^{-16}
C4 common-cell convergence (E16–E20)	0.00 (boolean)
C5 purpose (E21–E25)	0.00
C6 motivation heterogeneity (E26–E30)	1.81×10^{-16}
C7 cell- vs. point-meaning (E31–E35)	0.00 (boolean)
C8 Gödel-residue and bias (E36–E40)	0.00
C9 cell exteriority (E41–E45)	0.00 (boolean)
Aggregate (45)	1.89×10^{-16}

Table 1: Validation summary across 45 experiments. All verdicts PASS at machine precision. Maximum aggregate relative error 1.89×10^{-16} , within IEEE-754 double-precision rounding.

19 Connections to classical literature

19.1 Game theory and coordination

Classical game theory and the Schelling–Lewis–Aumann literature on coordination [42, 35, 4, 40, 51, 23] grounds coordination in iterated common knowledge of focal points or in equilibrium selection. Theorem 17.3 dispenses with the iterated hierarchy by placing the focal point in outcome space (the cell) rather than in agents’ representations. The closure properties (Theorem 5.6) recover the lattice structure familiar from coalition theory [23] without the common-knowledge prerequisite.

19.2 Distributed computing

The Fischer–Lynch–Paterson impossibility theorem [19] and the Byzantine generals problem [32] establish fundamental limits on distributed consensus under faulty agents. Our framework offers a complementary perspective: composite floor $S_b(E)$ measures the residual

“disagreement granularity” of a multi-agent system, and the catalytic composition law (Theorem 3.7) gives a quantitative reduction recipe. We do not prove FLP from scratch; we provide a calculus in which FLP-style impossibilities show up as positive composite-floor lower bounds.

19.3 Belief revision and epistemic logic

The AGM postulates for belief revision [1] and the Hintikka–Aumann–Halpern–Moses–Vardi framework for reasoning about knowledge [25, 5, 18] provide elaborate machinery for tracking individual beliefs and their interactions. Our framework treats belief as a derived quantity (Definition 4.1) of the receiver’s decoder–projection chain. Belief revision in our calculus corresponds to receiver substitution at fixed cell (?? from the foundations paper). The cell-truth framework subsumes the syntactic AGM machinery while abstracting away the propositional commitments.

19.4 Information theory and statistics

Shannon’s mutual-information chain rule [43, 17] and the Kullback–Leibler additivity over independent factors [31] mirror our catalytic composition law (Theorem 12.2): the dimensionless deficit $1 - S_b/\Sigma$ plays the role of a probability of success, and independent composition is multiplicative. Fisher information [20] and statistical hypothesis testing [34, 53] provide the standard measure-theoretic backdrop for the dispersion σ in methodologies.

19.5 Foundational logic

Gödel’s incompleteness theorems [21], Tarski’s undefinability theorem [47], the halting problem [49, 16], and Chaitin’s information-theoretic incompleteness [15] each establish a strictly positive floor on what a bounded formal system can decide. Our receiver floor β is the operational analogue: the same kind of strictly-positive lower bound, but receiver-relative and multiplicatively decomposable.

19.6 Functional analysis

Banach’s contraction principle [7] provides the fixed-point machinery for methodological floor (Theorem 8.1). Koopman operators [30] are structural analogues of the decoder–projection lift. Stone duality [46] suggests an alternative algebraic packaging of representations as Boolean algebras. We use these as analogues, not as premises.

19.7 Combinatorics and category theory

The partition encoding of Remark 2.11 connects to classical partition theory [2, 45]. The categorical formulation of purposes (Remark 7.6), the goal-quotient functor (Construction 12.1), and the OR-monoid structure (Proposition 12.3) live in standard categorical foundations [37, 6, 8].

19.8 Topology and measure

Reachability volumes (Definition 6.1) are measure-theoretic quantities; the underlying topology is supplied by d [27, 38]. Cell exteriority (Theorem 17.1) is a topological fact about $\mathcal{X}$ alone.

19.9 Statistical mechanics as analogue

Borel–Cantelli [13, 14], Boltzmann’s entropy counting [12], and the Landauer limit [33, 9] provide structural analogues for our asymptotic floor result (Corollary 3.9) and for the dimensional-deficit interpretation of β . We use them as analogues; no statistical-mechanical premise is assumed.

20 Conclusion

We have developed a comprehensive mathematical theory of multi-agent coordination from a single primitive — the receiver-relative scalar S measuring residual semantic distance to an action-cell. The principal results are:

- (1) *Cell-Truth foundations* (Section 2): the receiver calculus, the floor $\beta > 0$, the action-cell with positive tolerance, and representational invariance.

- (2) *Multi-agent algebra* (Section 3): the catalytic composition law for n -agent ensembles, reachability monotonicity (Corollary 3.8), and asymptotic floor reduction (Corollary 3.9).
- (3) *Belief Incompatibility and Common-Cell Convergence* (Sections 4 to 6): agents with mutually disjoint representation spaces nevertheless attain the same action-cell (Theorem 5.1); reachability volume satisfies a provable lower bound (Theorem 6.3).
- (4) *Purpose* (Sections 7 to 9): purpose formalized as a fixed-point cell of the multi-agent composite floor functional, with proven existence, uniqueness up to reachability equivalence, stability under perturbation, and an ω -limit characterization for dynamical agents.
- (5) *Motivation Heterogeneity* (Sections 10 to 12): the composite floor depends only on per-agent floors, not on goal-content; replication’s effectiveness derives from heterogeneity, not repetition; the goal-quotient yields a clean categorical structure with the OR-monoid as its codomain.
- (6) *Reformulation of classical impossibilities* (Sections 13 to 17): the eleven meaning-prerequisites collapse to the single condition $\beta = 0$; the Gödelian residue is exactly $\beta > 0$; the observation–bias equivalence is the decoder–projection composition; reality-independence is cell exteriority; and coordination requires no common knowledge.

The calculus is independent, self-contained, and validated to machine precision across forty-five numerical experiments. It provides a unified framework for reasoning about multi-agent coordination, purpose, and the relationship between bounded inquiry and operational truth without requiring shared knowledge, shared concepts, shared motivation, or shared methodology. What is shared is the *action-cell*: a structure in outcome space, not in any receiver’s representation. The cell is the coordination object; agents need only decode-to-it.

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